

By analyzing f' (and also f'') we can figure out a lot of information about the original function f . In this section we will use f' and f'' to sketch the original function.

The first derivative

We know the slope of a graph at any point is given by $f'(x)$. Positive slope means increasing, negative means decreasing.

Just to state it

Increasing/Decreasing test

a) If $f'(x) > 0$ on an interval, then f is increasing on that interval

b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval

Ex: Find where $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and decreasing.

$$f'(x) = 12x^3 - 12x^2 - 24x$$

$$= 12x(x^2 - x - 2)$$

$$= 12x(x-2)(x+1) \quad \sim \text{Try to write as a Product}$$

Find where $f'(x) = 0$ or DNE

$$f'(x) = 0 \text{ when } x = 0, 2, -1$$

This gives intervals, $(-\infty, -1)$, $(-1, 0)$, $(0, 2)$, $(2, \infty)$

	$12x$	$x-2$	$x+1$	$f'(x)$	$f(x)$
$(-\infty, -1)$	-	-	-	-	dec
$(-1, 0)$	-	-	+	+	inc
$(0, 2)$	+	-	+	-	dec
$(2, \infty)$	+	+	+	+	inc

So, f is increasing on $(-1, 0) \cup (2, \infty)$
decreasing on $(-\infty, -1) \cup (0, 2)$

The notion of increasing and decreasing allows us to characterize local maximum and minimum values.

The first derivative test

Suppose c is a critical number of a continuous function f ($f'(c)=0$ or DNE)

a) If f' changes from positive to negative at c , then f has a local maximum at c .

b) If f' changes from negative to positive at c , then f has a local minimum at c .

c) If f' does not change signs at c , then $f(c)$ is not a local extrema.

In the above example:

f has a local max at $x=0$, and
 f has a local min at $x=-1, 2$

Ex: Find the local maximum and minimum values of

$$g(x) = x + 2\sin(x) \quad \text{for} \quad 0 \leq x \leq 2\pi.$$

First find critical numbers.

$$g'(x) = 1 + 2\cos(x)$$

$$g'(x) = 0 \Rightarrow 1 + 2\cos(x) \Rightarrow \cos(x) = -\frac{1}{2} \Rightarrow \underline{x = \frac{2\pi}{3}, \frac{4\pi}{3}}$$

This gives intervals $(0, \frac{2\pi}{3})$, $(\frac{2\pi}{3}, \frac{4\pi}{3})$, $(\frac{4\pi}{3}, 2\pi)$

	$1 + 2\cos(x)$	$f(x)$
$(0, \frac{2\pi}{3})$	+	inc
$(\frac{2\pi}{3}, \frac{4\pi}{3})$	-	dec
$(\frac{4\pi}{3}, 2\pi)$	+	inc

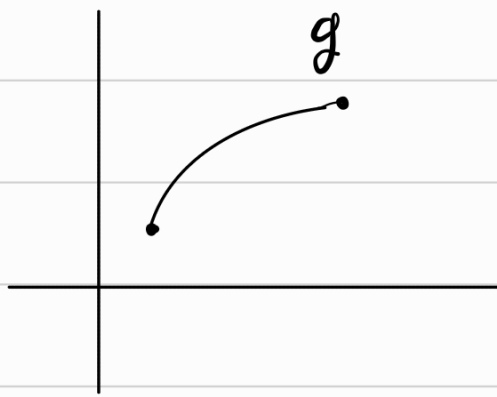
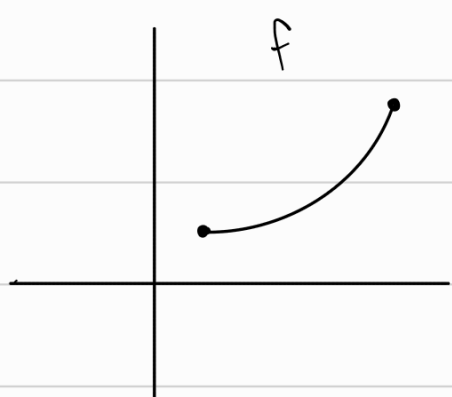
So $x = \frac{2\pi}{3}$ is a maximum, $f(\frac{2\pi}{3}) = \frac{2\pi}{3} + \sqrt{3}$

$x = \frac{4\pi}{3}$ is a minimum, $f(\frac{4\pi}{3}) = \frac{4\pi}{3} - \sqrt{3}$

The Second Derivative

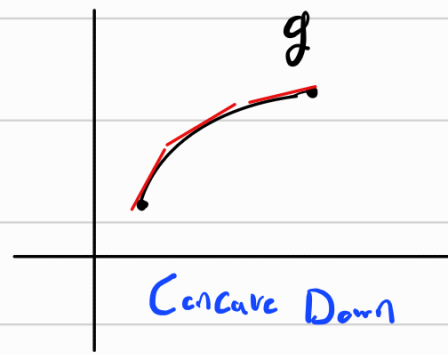
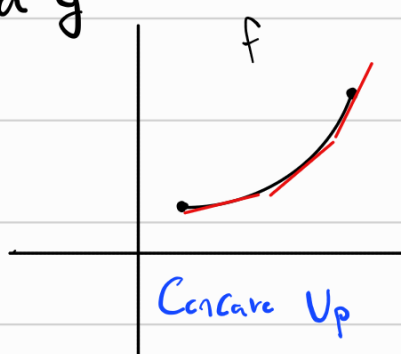
The first derivative tells us a lot about the function but not everything. More information comes from the second derivative. What type of information?

Consider the two functions



In both cases $f'(x) > 0$ and $g'(x) > 0$,
So if all we knew about a function was that its derivative is positive, we still can't tell the general shape.

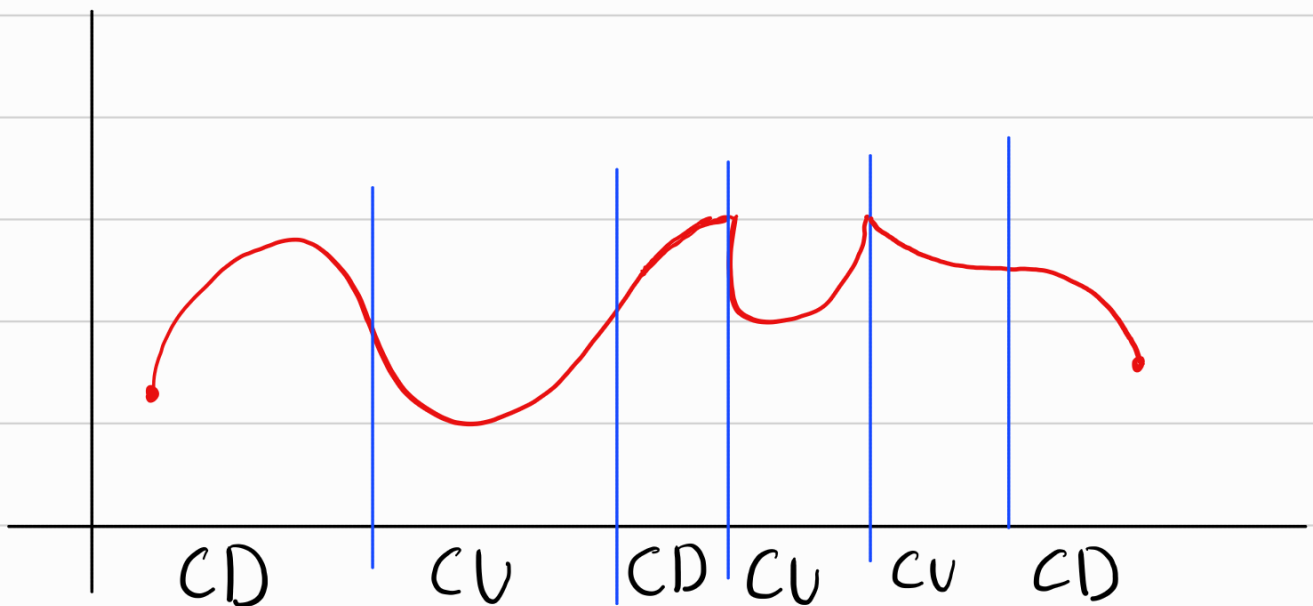
For this, we have a notion of Concavity. Observe the tangents to f and g



For f , all tangents fall below the curve. For g , all tangents are above the curve.

Def:

- If the graph f lies above all of its tangents on an interval I , then f is called Concave upward on I .
- If the graph f lies below all of its tangents on an interval I , then f is called Concave downward on I .



To analytically classify concavity, we can use the second derivative.

Concavity Test

a. If $f''(x) > 0$ on an interval I , then f is
Concave UP on I
 $\hookrightarrow CU$

b. If $f''(x) < 0$ on an interval I , then f is
Concave Down on I .
 $\hookrightarrow CD$

Def:

A point P on a curve $y = f(x)$ is called an inflection point if f is continuous there and the curve changes Concavity type there.

Since Concavity relates to the second derivative, inflection points are where f'' changes sign. More specifically,

If the point $(c, f(c))$ is an inflection point,
 $f''(c) = 0$ or DNE.

Like with max and mins, the statement doesn't work

backward If $f''(c) = 0$ or DNE, the point might not be an Inflection point. So where $f''(x) = 0$ or DNE are only Potential Inflection Points, we must check there is a change in Concavity.

² Consider $f(x) = x^4$

Ex: Find the Inflection Points of
 $f(x) = x^4 - 12x^3 + 84x^2 + 4$

$$f'(x) = 4x^3 - 36x^2 + 84x$$

$$f''(x) = 12x^2 - 72x = 12x(x-6)$$

$$f''(x) = 0 \rightarrow x=0 \text{ or } x=6$$

We still need to check change in Concavity

	$12x$	$x-6$	$f''(x)$	Concavity
$(-\infty, 0)$	-	-	+	Concave UP
$(0, 6)$	+	-	-	Concave Down
$(6, \infty)$	+	+	+	Concave UP

So $x=0$ and $x=6$ are inflection Points.

Ex: Find Inflection Points of
 $f(x) = x(x+4)^{2/3}$

$$f'(x) = (x+4)^{2/3} + \frac{2}{3}x(x+4)^{-1/3}$$

$$f''(x) = \frac{2}{3}(x+4)^{-1/3} + \frac{2}{3}(x+4)^{-1/3} - \frac{2}{9}x(x+4)^{-4/3}$$

$$= \frac{4}{3} \frac{x+4}{(x+4)^{4/3}} - \frac{2x}{9(x+4)^{4/3}}$$

$$= \frac{12x+48}{9(x+4)^{4/3}} - \frac{2x}{9(x+4)^{4/3}}$$

$$= \frac{10x+48}{9(x+4)^{4/3}}$$

$$f''(x) = 0 \text{ if } x = -\frac{24}{5}, \quad f''(x) \text{ DNE if } x = -4$$

	$10x+48$	$9(x+4)^{4/3}$	$f''(x)$	Concavity
$(-\infty, -\frac{24}{5})$	-	+	-	Down
$(-\frac{24}{5}, -4)$	+	+	+	UP
$(-4, \infty)$	+	+	+	UP

So $x = -\frac{24}{5}$ is inflection point

Ex: Sketch a graph such that

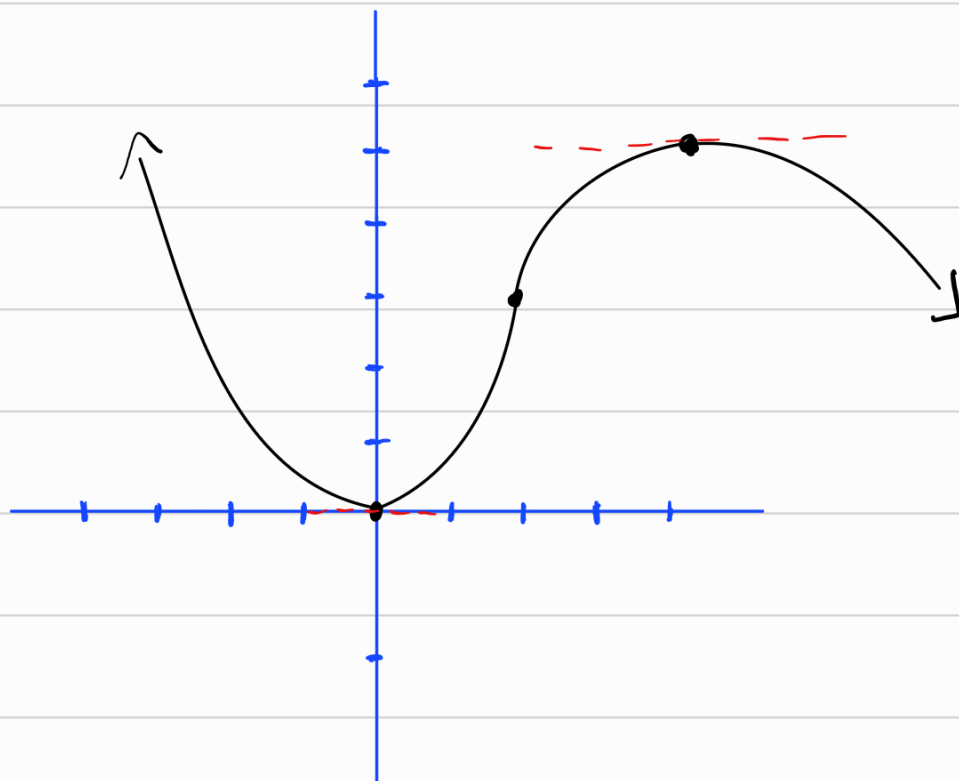
i) $f(0)=0$, $f(2)=3$, $f(4)=6$

ii) $f'(0)=0$, $f'(4)=0$

iii) $f'(x) > 0$ for $0 < x < 4$, $f'(x) < 0$ for $x < 0$, $x > 4$

iv) $f''(x) > 0$ for $x < 2$, $f''(x) < 0$ for $x > 2$

- Start by fixing points from (i)
- Notice $(0,0)$ and $(4,6)$ are min and max as the derivative changes sign
- Look at Concavity



We can use properties of concavity to test for local extrema.

The general idea of concavity was that the curve would "bend" up or down. If we considered a point that could be a local extrema, if the curve was concave down there (bending down) then it should be a maximum. Likewise for minimums.



This gives us the following Test

Second Derivative Test

Suppose f'' is continuous near c

- If $f'(c) = 0$ and $\underline{f''(c) > 0}$, then f has a local minimum at c .

- If $f'(c) = 0$ and $\underline{f''(c) < 0}$, then f has a local maximum at c .

Ex: Find local extrema of $f(x) = x^4 - 4x^3$

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$$

$$f''(x) = 12x^2 - 24x$$

$$f'(x) = 0 \text{ if } x=0 \text{ or } x=3$$

$$f''(3) = 12(9) - 24(3) = 36 > 0 \Rightarrow \text{local min}$$

$$f''(0) = 0 \Rightarrow \text{Second derivative test inconclusive}$$

$\hookrightarrow f'(x)$ doesn't change sign, so this isn't max nor min

f has local minimum at $x=3$

f has no local maximum

Sketching Graphs:

We can now find local extrema, points of inflection, intervals of inc/dec, and intervals of concavity. This allows us to make a very good sketch of a curve.

Ex: Sketch $f(x) = x^{2/3}(6-x)^{1/3}$

$$f'(x) = \frac{2}{3}x^{-1/3}(6-x)^{1/3} - \frac{1}{3}x^{2/3}(6-x)^{-2/3}$$

$$= \frac{2(6-x)^{1/3}}{3x^{1/3}} \cdot \frac{(6-x)^{2/3}}{(6-x)^{2/3}} - \frac{x^{2/3}}{(6-x)^{2/3}} \cdot \frac{x^{1/3}}{x^{1/3}}$$

$$= \frac{2(6-x) - x}{3x^{1/3}(6-x)^{2/3}}$$

$$= \frac{12-3x}{3x^{1/3}(6-x)^{2/3}}$$

$$\underline{f'(x) = \frac{4-x}{x^{1/3}(6-x)^{2/3}}}$$

$$f''(x) = \frac{-x^{1/3}(6-x)^{2/3} - (4-x)\left(\frac{1}{3}x^{-2/3}(6-x)^{2/3} - \frac{2}{3}x^{1/3}(6-x)^{-1/3}\right)}{x^{2/3}(6-x)^{4/3}}$$

$$= \frac{-x^{1/3}(6-x)^{2/3} \frac{3x^{2/3}(6-x)^{1/3}}{3x^{2/3}(6-x)^{1/3}} - \frac{(4-x)(6-x)^{2/3}}{3x^{2/3}} \frac{(6-x)^{1/3}}{(6-x)^{1/3}} + \frac{2(4-x)x^{1/3}}{3(6-x)^{1/3}} \frac{x^{2/3}}{x^{2/3}}}{x^{2/3}(6-x)^{4/3}}$$

$$= \frac{\frac{-3x(6-x)}{3x^{2/3}(6-x)^{1/3}} - \frac{24-10x+x^2}{3x^{2/3}(6-x)^{1/3}} + \frac{2(4-x)x}{3(6-x)^{1/3}x^{2/3}}}{x^{2/3}(6-x)^{4/3}}$$

$$= \frac{\underline{-18x+3x^2} - \underline{24+10x-x^2} + \underline{8x-2x^2}}{3x^{4/3}(6-x)^{5/3}}$$

Combine
Denominators

$$= \frac{-24}{3x^{4/3}(6-x)^{5/3}}$$

$$f''(x) = \frac{-8}{x^{4/3}(6-x)^{5/3}}$$

All the annoying work is out of the way, let's analyze the derivatives

$$f'(x) = \frac{4-x}{x^{1/3}(6-x)^{2/3}}, \quad f''(x) = \frac{-8}{x^{4/3}(6-x)^{5/3}}$$

1. Increasing/Decreasing

$$f'(x) = 0 \quad \text{if } \underline{x=4}$$

$$f'(x) \text{ DNE} \quad \text{if } \underline{x=0} \text{ or } \underline{x=6}$$

	$4-x$	$x^{1/3}$	$(6-x)^{2/3}$	$f'(x)$	Inc/dec
$(-\infty, 0)$	+	-	+	-	dec
$(0, 4)$	+	+	+	+	inc
$(4, 6)$	-	+	+	-	dec
$(6, \infty)$	-	+	+	-	dec

f has max at $x=4$, $f(4)=2^{5/3}$
 min at $x=0$, $f(0)=0$

2. Concavity

$$f''(x) = 0 \quad \text{never}$$

$$f''(x) \text{ DNE} \quad \text{if } x=0 \text{ or } x=6$$

	-8	$x^{4/3}$	$(6-x)^{5/3}$	$f''(x)$	Concavity
$(-\infty, 0)$	-	+	+	-	Down CD
$(0, 6)$	-	+	+	-	Down CD
$(6, \infty)$	-	+	-	+	Up CU

So there is a point of inflection at $x=6$, $f(6)=0$

3. Graph

Notice $f(x)=0$ if $x=0$ or $x=6$. This will help graph.

$$f(x) = x^{2/3}(x-6)^{1/3}$$

